

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Scenario 1: Manufacturing Quality Inspection

Introduction

In modern manufacturing industries, automated quality inspection systems play a vital role in ensuring that products meet required quality standards. These systems use digital cameras and image processing techniques to inspect metal components for defects such as cracks, scratches, dents, and surface irregularities. The accuracy of these systems depends on the quality of the captured images.

In this scenario, the inspection system captures images of metal components, but electrical interference introduces repetitive horizontal stripe patterns across the images. This degradation reduces image quality and makes automatic defect detection unreliable. To restore image quality, an appropriate image enhancement technique must be selected. The following sections explain the problem, its impact, and the most suitable solution.

(a) Interpretation of the Scenario and Identification of the Noise

The given scenario describes an automated inspection system where images are affected by repetitive horizontal stripes due to electrical interference. This type of degradation is known as **Periodic Noise**.

Periodic noise appears as regular repeating patterns such as horizontal or vertical stripes. It is commonly caused by electrical disturbances, electromagnetic interference, faulty sensors, or power supply fluctuations during image acquisition. Unlike random noise, periodic noise follows a regular pattern and occupies specific frequency locations in the frequency domain.

In this case, the horizontal stripes indicate that the image has been corrupted by periodic interference, making it difficult for the inspection system to analyze the actual surface of the metal component accurately.

Type of Noise:

Periodic Noise (caused by electrical interference).

(b) Effect of Image Degradation on Inspection Accuracy

Image degradation significantly affects the performance of an automated quality inspection system. Since the system depends on image clarity for detecting defects, the presence of horizontal stripe noise creates several problems.

Effects include:

- The repetitive stripes hide small defects such as scratches, cracks, and dents.
- Important surface details become distorted.
- Image contrast decreases, making feature extraction difficult.
- Edge detection algorithms produce incorrect results.
- The inspection system may identify defects where none exist (False Positives).
- Actual defects may remain undetected (False Negatives).
- Product quality assessment becomes unreliable.
- Manufacturing efficiency decreases because defective products may pass inspection or good products may be rejected.

(c) Recommended Enhancement Technique

The most suitable enhancement technique is the Frequency Domain Notch Filter.

Since periodic noise appears at specific frequencies, it is easier to remove in the frequency domain than in the spatial domain.

Steps involved:

Step 1: Capture the noisy image.

Step 2: Apply the Fast Fourier Transform (FFT) to convert the image from the spatial domain into the frequency domain.

Step 3: Observe the Fourier spectrum and identify the bright peaks representing periodic noise.

Step 4: Apply a Notch Filter to suppress only the unwanted frequency components.

Step 5: Apply the Inverse Fast Fourier Transform (IFFT) to convert the filtered image back to the spatial domain.

(d) Justification for Choosing Frequency Domain Enhancement

The frequency-domain approach is more suitable because periodic noise occupies distinct frequency locations in the Fourier spectrum. A notch filter removes only those unwanted frequencies without affecting the rest of the image.

Advantages of Frequency-Domain Notch Filter

- Removes periodic noise effectively.
- Preserves edges and fine surface details.
- Improves defect visibility.
- Provides higher inspection accuracy.
- Does not blur the image significantly.
- Suitable for repetitive interference patterns.

Why Spatial-Domain Methods Are Less Suitable

Spatial-domain filters such as the Mean Filter, Median Filter, and Gaussian Filter are mainly designed for random noise like Gaussian noise or Salt-and-Pepper noise.

Their limitations include:

- They cannot selectively remove periodic stripe patterns.
- They blur image edges.
- Fine defects may disappear during filtering.
- Inspection accuracy decreases because important details are lost.

Therefore, a frequency-domain notch filter is the preferred solution for removing repetitive electrical interference.

(e) Logical Sequence of the Solution

The complete enhancement process can be carried out in the following logical sequence:

1. Capture the image of the metal component.
2. Detect the presence of horizontal stripe interference.
3. Convert the image into the frequency domain using the Fast Fourier Transform (FFT).
4. Locate the periodic noise frequencies in the Fourier spectrum.

5. Apply a Notch Filter to remove only the unwanted frequencies.
6. Perform the Inverse Fast Fourier Transform (IFFT) to reconstruct the enhanced image.
7. Use the enhanced image for automatic defect detection.
8. Generate an accurate quality inspection report.

Conclusion

The repetitive horizontal stripes observed in the captured images are caused by periodic noise resulting from electrical interference. This noise reduces image quality, hides important surface defects, and decreases the reliability of automated quality inspection.

Scenario 2: Autonomous Vehicle Navigation

Introduction

Autonomous vehicles use cameras and image processing techniques to identify road lanes, road edges, traffic signs, pedestrians, and obstacles for safe navigation. The accuracy of these systems depends on the quality of the captured images. During heavy rain, raindrops, water splashes, and poor visibility introduce noise into the images, making it difficult to detect important road features.

In this scenario, although the captured images contain noise, the lane markings and road edges must remain sharp because they are essential for steering, lane keeping, and collision avoidance. Therefore, an image enhancement technique is required that removes noise while preserving edge information.

(a) Interpretation of the Scenario and the Challenge Involved

The scenario describes a self-driving car operating during heavy rainfall. Rain introduces image noise and reduces visibility, affecting the camera's ability to capture clear road images.

The major challenge is to remove the noise without damaging important road features, especially lane markings and road boundaries. If these edges become blurred, the autonomous vehicle may fail to recognize lanes correctly, increasing the risk of navigation errors and accidents.

Thus, the system requires an edge-preserving filtering technique that improves image quality while maintaining critical structural details.

(b) Importance of Preserving Edges While Removing Noise

Edges represent the boundaries of important objects in an image. In autonomous driving, lane markings, road edges, vehicles, pedestrians, and traffic signs are identified based on their edges.

Importance of edge preservation:

- Lane markings remain clearly visible.
- Road boundaries can be accurately detected.
- Object detection algorithms perform more accurately.
- Vehicle steering decisions become reliable.
- Collision avoidance systems function effectively.
- Navigation accuracy improves even during bad weather.
- Safety of passengers and pedestrians is enhanced.

(c) Recommended Filtering Technique

The most suitable filtering technique is the Bilateral Filter.

The Bilateral Filter removes image noise while preserving sharp edges by considering both:

- Spatial distance between neighbouring pixels.
- Intensity difference between neighbouring pixels.

Unlike traditional smoothing filters, it smooths only similar pixels and avoids averaging across edges. As a result, the filter effectively reduces rain-induced noise while keeping lane markings and road edges sharp.

(d) Justification for Selecting the Bilateral Filter

The Bilateral Filter is preferred because it is specifically designed for **edge-preserving** smoothing.

Advantages of Bilateral Filter

- Removes rain-induced image noise effectively.
- Preserves lane markings and road edges.
- Maintains object boundaries.
- Reduces image blur.
- Improves feature extraction.
- Enhances navigation accuracy.
- Produces better results than ordinary smoothing filters.

(e) Logical Sequence of the Solution

The image enhancement process can be carried out in the following systematic sequence:

1. Capture the road image using the vehicle camera.
2. Detect the presence of rain-induced noise.
3. Apply the Bilateral Filter to smooth noisy regions.
4. Preserve lane markings and road edges during filtering.
5. Perform edge and lane detection on the enhanced image.
6. Identify road boundaries and obstacles accurately.
7. Make safe steering and navigation decisions.
8. Continue autonomous driving with improved reliability.

Conclusion

Heavy rain introduces noise into images captured by autonomous vehicles, making road analysis more difficult. However, preserving lane markings and road edges is essential for safe navigation. The Bilateral Filter is the most suitable filtering technique because it effectively removes noise while maintaining sharp edges and important image details. Compared to conventional filters such as Mean, Gaussian, and Median filters, the Bilateral Filter provides superior edge preservation, resulting in more accurate lane detection, improved navigation, and enhanced road safety.